A phase I study investigating the safety and pharmacokinetics of highly bioavailable curcumin (Theracurmin) in cancer patients.

BACKGROUND: A growing number of preclinical studies have demonstrated that curcumin could be a promising anticancer drug; however, poor bioavailability has been the major obstacle for its clinical application. To overcome this problem, we developed a new form of curcumin (Theracurmin) and reported high plasma curcumin levels could be safely achieved after a single administration of Theracurmin in healthy volunteers. In this study, we aimed to evaluate the safety of repetitive administration of Theracurmin in cancer patients. METHODS: Pancreatic or biliary tract cancer patients who failed standard chemotherapy were eligible for this study. Based on our previous pharmacokinetic study, we selected Theracurmin containing 200 mg of curcumin (Level 1) as a starting dose, and the dose was safely escalated to Level 2, which contained 400 mg of curcumin. Theracurmin was orally administered every day with standard gemcitabine-based chemotherapy. In addition to safety and pharmacokinetics data, NF-κB activity, cytokine levels, efficacy, and quality-of-life score were evaluated. RESULTS: Ten patients were assigned to level 1 and six were to level 2. Peak plasma curcumin levels (median) after Theracurmin administration were 324 ng/mL (range, 47-1,029 ng/mL) at Level 1 and 440 ng/mL (range, 179-1,380 ng/mL) at Level 2. No unexpected adverse events were observed and 3 patients safely continued Theracurmin administration for >9 months. CONCLUSIONS: Repetitive systemic exposure to high concentrations of curcumin achieved by Theracurmin did not increase the incidence of adverse events in cancer patients receiving gemcitabine-based chemotherapy.